

EXPERIMENT 7

COLUMNAR TRANSPOSITION CIPHER

AIM

To implement the Columnar Transposition Cipher using C.

ALGORITHM

1. Read the plaintext and keyword.
2. Arrange the plaintext row-wise in a matrix.
3. Number the columns according to the alphabetical order of the keyword.
4. Read the columns in sorted-key order.
5. Display the encrypted text.

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char text[100];
    int cols, len, rows, i, j, k=0;

    printf("Enter Plain Text: ");
    scanf("%s", text);

    printf("Enter Number of Columns: ");
    scanf("%d", &cols);

    len = strlen(text);
    rows = (len + cols - 1) / cols;

    char mat[20][20];

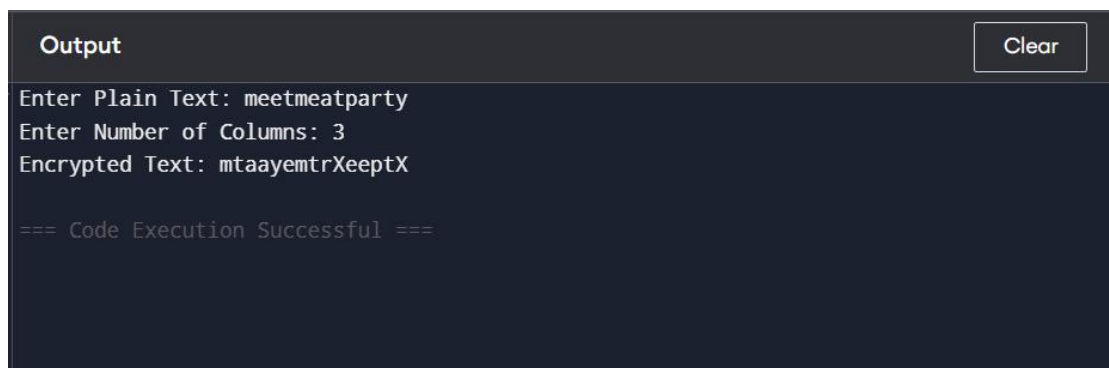
    k = 0;
    for(i=0; i<rows; i++)
        for(j=0; j<cols; j++)
            if(k<len)
                mat[i][j]=text[k++];
            else
                mat[i][j]='X';

    printf("Encrypted Text: ");
```

```
    for(j=0;j<cols;j++)
        for(i=0;i<rows;i++)
            printf("%c",mat[i][j]);

    return 0;
}
```

OUTPUT



The screenshot shows a dark-themed window titled "Output" with a "Clear" button in the top right corner. The text inside the window displays the input and output of the program: "Enter Plain Text: meetmeatparty", "Enter Number of Columns: 3", and "Encrypted Text: mtaayemtrXeeptX". At the bottom, it states "=== Code Execution Successful ===".

```
Output Clear
Enter Plain Text: meetmeatparty
Enter Number of Columns: 3
Encrypted Text: mtaayemtrXeeptX

=== Code Execution Successful ===
```

RESULT

Thus the Columnar Transposition Cipher was implemented successfully using C.